

Mohamed Felhi

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PROFESSIONAL SUMMARY

IT Engineering student at ESPRIT with practical experience in software development, robotics, embedded systems, and database-driven applications. Comfortable working across web development, desktop applications, game development, and intelligent robotic systems, with strong hands-on involvement in both hardware and software team projects.

EDUCATION

ESPRIT School of Engineering, Ariana, Tunisia | IT Engineering Student

Currently in the 2nd year of the engineering cycle. Studied C, C++, Python, Qt, HTML, CSS, JavaScript, PHP, MySQL, Oracle Database, and introductory Node.js and Three.js.

Bech Hamba High School, Bizerte, Tunisia | Baccalaureate in Computer Science

INTERSHIP EXPERIENCE

STEG – Tunisian Company of Electricity and Gas | Human and Social Training Intern, Customer Service Department

- Welcomed customers at the service desk and responded to their requests.
- Assisted with the administrative organization of bills by sorting files and verifying references.
- Developed communication, organization, and teamwork skills while discovering how a public utility service operates.

ACADEMIC PROJECTS

2D Survival Game | C, SDL 1.2, MySQL

- Built a 2D adventure-survival game in which stranded friends must be rescued from enemies on an island.
- Developed major gameplay systems including the player, enemies, animals, projectiles, platforms, UI, puzzles, and chest rewards.
- Implemented two playable styles: sword-based melee combat and bow-based ranged combat.
- Added gameplay effects such as regeneration, stamina recovery, strength boosts, and arrow collection.
- Created movement, jump, combat, and fire-damage animations.
- Added local multiplayer, used Git for version control, and connected saved progress to a MySQL database.
- Used Adobe Illustrator to edit and create visual assets such as backgrounds and character elements.

Smart Driving School Application | C++, Qt, Oracle, Ollama

- Developed the client management module for a smart driving school desktop application.
- Built the admin interface for adding, deleting, modifying, sorting, and searching client records.
- Implemented automatic email suggestions for the next driving session after a completed session.
- Developed a recommendation feature that predicts exam timing, remaining hours, frequent mistakes, and personalized guidance using previous-session data and an LLM workflow through Ollama.
- Used Git for version control during development.

ROBOTICS AND LEADERSHIP EXPERIENCE

IEEE Member – RAS and CIS Chapters | Robotics and Technical Activities

- Active IEEE member, with stronger involvement in the Robotics and Automation Society (RAS) chapter.
- Participated in workshops, supported technical projects, and helped with robotics event organization.
- Contributed as an organizer for activities related to competitions such as RAS ROBOTS.

Line Follower Competitions | Team Hardware Contributor

- Participated in line follower competitions such as Robots League 2.0 at the Higher School of Science and Technology of Hammam Sousse.

- Worked as the hardware-focused member of the team on the robot El Fras.
- Started with basic line follower methods using simple IR detection modules, then upgraded the robot using PID control and Pololu QTR-8A sensors.
- Added Bluetooth control to start and stop the robot and tune PID values directly from a phone without needing a PC.

PERSONAL ENGINEERING PROJECT

Cero – Autonomous Hexapod Robot | Python, C/C++, Raspberry Pi, Embedded Systems

- Currently developing an autonomous hexapod robot designed to walk, interact with its environment, and execute behavior commands.
- Project goal: combine mechanical design, embedded systems, and artificial intelligence in a single robotic platform.
- Main architecture: Raspberry Pi 5 for high-level AI and planning, two Raspberry Pi Pico boards for the six legs, and one Teensy for a 6-DOF robotic arm with gripper.
- Software split: Python on the Raspberry Pi for high-level coordination and C/C++ on microcontrollers for inverse kinematics and real-time motor control.
- Uses PCA9685 servo drivers, I2C communication, a 3S LiPo power system, voltage regulation, and fuse-based protection.
- Future work includes reinforcement learning for autonomous walking, vision-based navigation, natural language command interpretation, and improved gait stability.

TECHNICAL SKILLS

Programming	C, C++, Python, JavaScript, PHP, SQL
Web	HTML, CSS, JavaScript, PHP, basic Node.js, basic Three.js
Databases	MySQL, Oracle Database
Frameworks & Tools	Git, Qt, SDL 1.2, Ollama, Adobe Illustrator
Robotics & Embedded	Raspberry Pi 5, Raspberry Pi Pico, Teensy, I2C, PCA9685, PID control, robotics prototyping
Platforms	Ubuntu (virtual machine environment)

LANGUAGES

English – B2 | Dutch – A1

ADDITIONAL INFORMATION

- Interested in robotics, artificial intelligence, embedded systems, and intelligent automation.
- Experienced in both software and hardware project work.